

SOFTWARE ENGINEERING INTRODUCTION

Software Engineering Seminar

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Software Development
- 2 Object-Oriented Design
- 3 Domain-Driven Design
- 4 Software Methodologies
- 5 Information Systems



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Basics of Software Development I

- The **main idea** is to **solve real-world** problems using **software solutions**. One of the **main challenges** is the **complexity** of **systems**, and learning how to **manage** it.
- It is **not just** about writing **code**; you must keep the entire **software life cycle** in mind. This means thinking about *design*, *testing*, *deployment*, *maintenance*, and many other tasks.

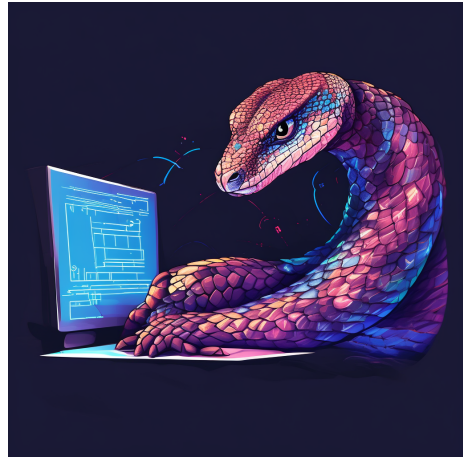


Figure: Prompt: Draw a python developer.



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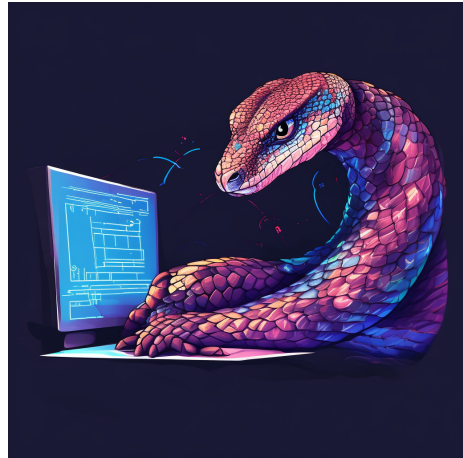


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Basics of Software Development II

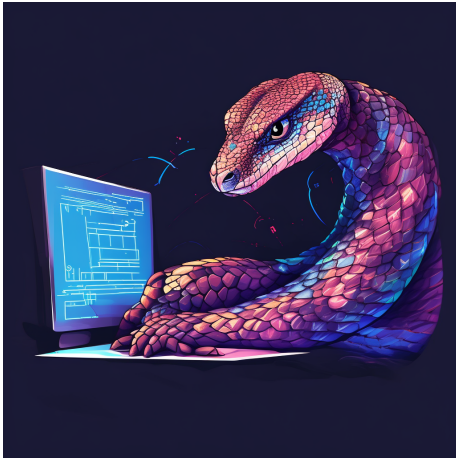


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- However, writing code is the most important task, and it is the main skill to have. You can write code to automate tests, deployments, integrations, and more.
- It is also vital to know a lot about software design, to propose good solutions, and to read every day in order to choose and use the best tools.
This is a crazy world.



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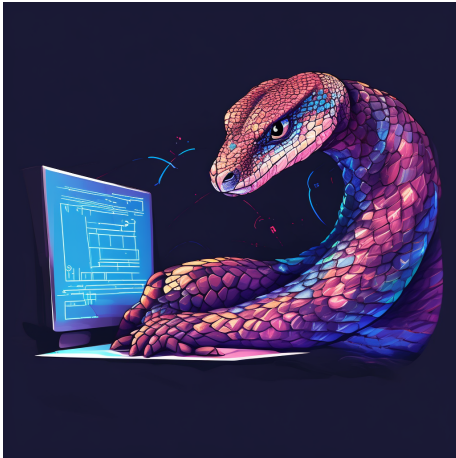
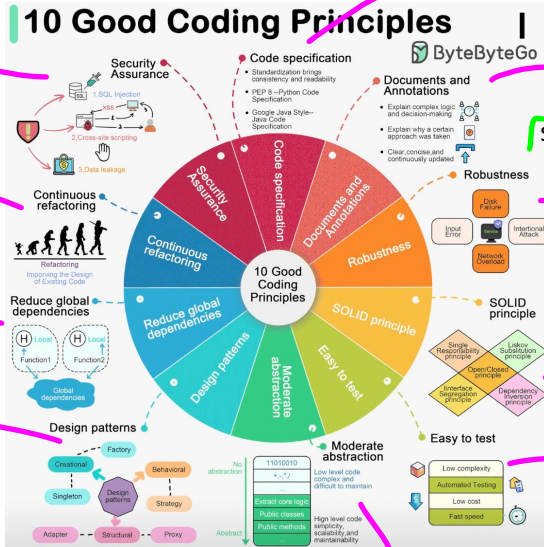


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10 Good Coding Principles



ByteByteGo

camelStyle

snake style



DataOps Vs. DevOps Vs. MLOps

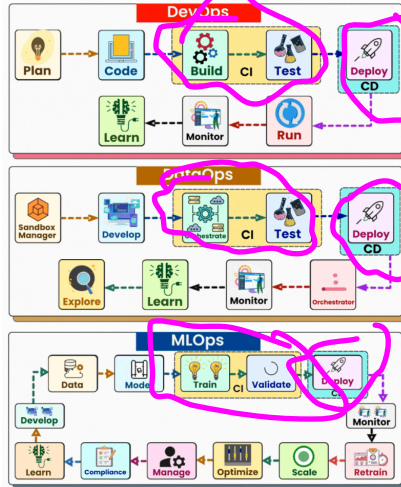


Brij Kishore Pandey



Save For Later

DevOps vs DataOps vs MLOps



Basics of Software Architecture I

- It is important to **develop** innovative and sophisticated software to provide effective solutions for **end users' needs**.
- **Software architecture** brings innovation and a robust structure.
- The goal of software architecture is to minimize the human effort required to build and maintain the expected system.



Figure: Prompt: A python developer watching a building architecture draws.



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Basics of Software Architecture II



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- A **software architecture** is the **skeleton** for a complete **software system**. It leads the **system** to be **scalable**, **reliable**, and **maintainable**. Also it helps to take better **technical decisions**.
- There are some **software architecture styles**, each one with **pros/cons**, and **specific use cases**. However, they try to provide a **reference solution** for a high-level structure of a software system.



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Types of Software Products

- **System Software:** Operating systems, device drivers, and utility programs.
- **Application Software:** Programs that perform specific user-oriented tasks (e.g., office suites, mobile apps).
- **Middleware:** Software that connects disparate systems and facilitates communication.
- **Embedded Software:** Specialized software designed to operate hardware in devices.
- **Enterprise Software:** Large-scale solutions like ERP, CRM, or SCM systems supporting *business operations*.



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Basics of Object-Oriented Design I

- **Object-oriented programming** has become one of the **most traditional** and popular **paradigms** in **software development**.
- It is based on the concept of **objects**, which can contain data in the form of **fields** (often known as *attributes* or *properties*) and code in the form of **procedures** (often known as *methods*).

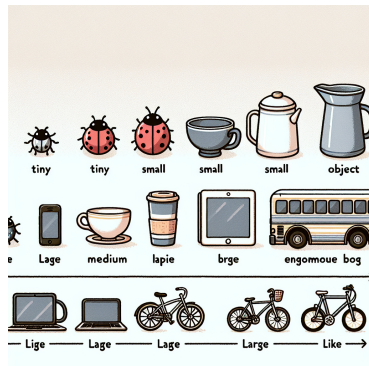


Figure: Prompt: Make an image of different real-world objects with binary inside each one.



Basics of Object-Oriented Design II

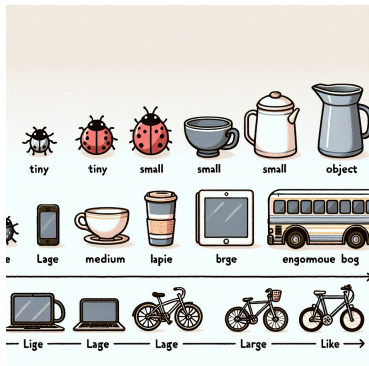


Figure: Prompt: Make an image of different real-world objects with binary inside each one.



- The idea is to design a **system modularly** to make it **easier** to **maintain** and understand. The idea is also to **emphasize** the **reuse of code**.
- The main principles of OOD are:

- Encapsulation
- Abstraction
- Inheritance
- Polymorphism



Basics of Object-Oriented Design II

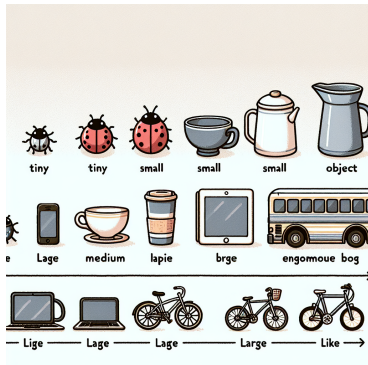


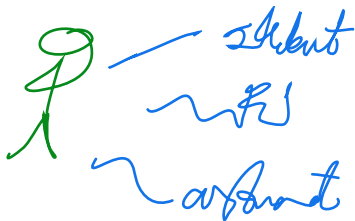
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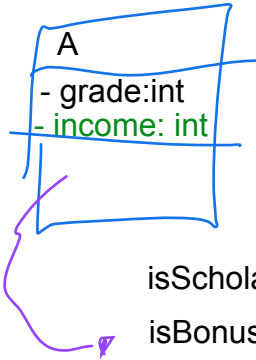
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Abstraction in OOD



Encapsulation in OOD



isScholarship(double min)

bool

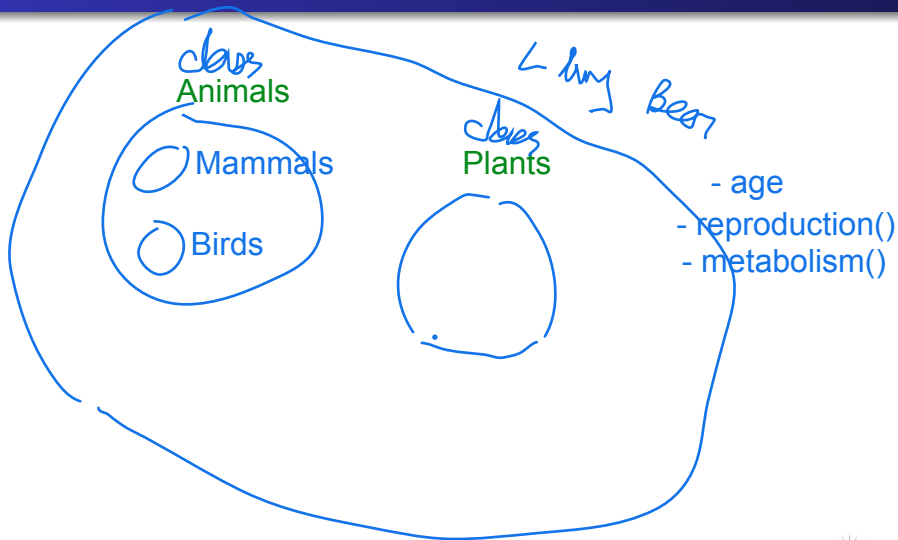
isBonus()

bool

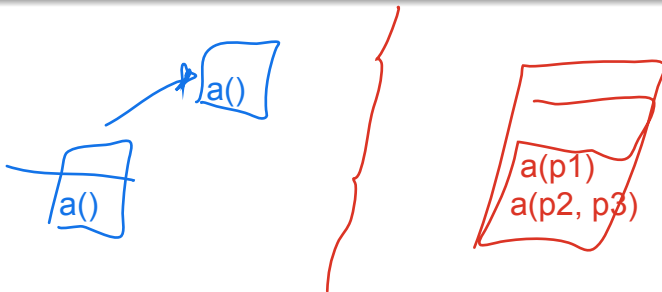
is.....() bool



Inheritance in OOD



Polymorphism in OOD

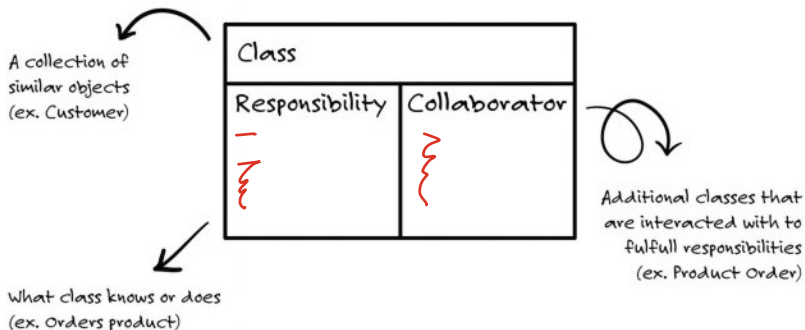


signature -> object — creation, public method
 method — name, parameters, return



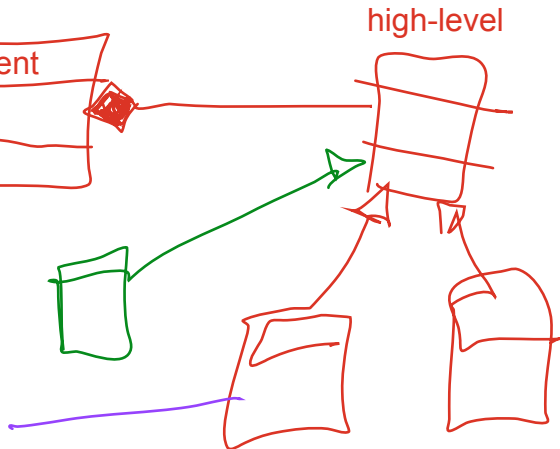
Class-Responsability-Collaboration Cards (CRC)

The **CRC cards** are a **brainstorming tool** used in the **design** of **object-oriented software**.



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Basics of Domain-Driven Design I

- **DDD** focuses on the **core domain** and **domain logic**. It is a way of **thinking** aimed at accelerating software projects that have to **deal with complicated domains**.
- The essential terms of **DDD** are *context, model, ubiquitous language, bounded context, and business logic in layers*.
- **DDD** is a set of **principles** and **patterns** that help **design** a system to ensure alignment with **real-world business needs**.

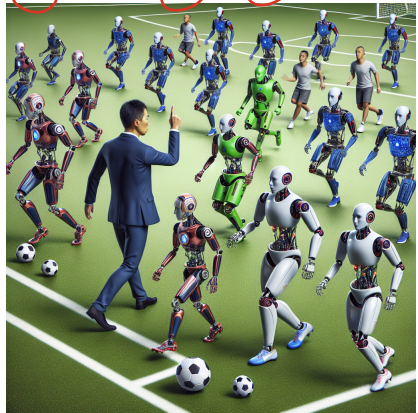


Figure: Prompt: Draw a soccer coach teaching robots soccer players.



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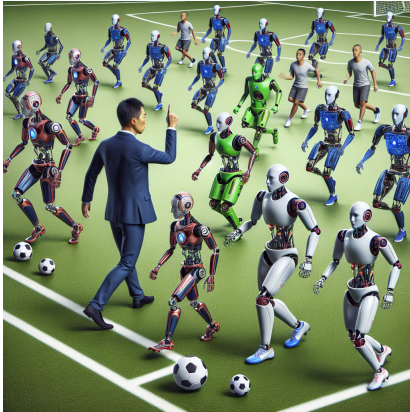


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- The **main principles** of **DDD** are:
 - Focus on the **core domain**.
 - Base complex designs on models of the domain.
 - Constantly collaborate with domain experts.
 - Develop a knowledge-rich model.
- The **business logic** in layers is showed as follows:

Domain Logic
Application Logic
Presentation Logic



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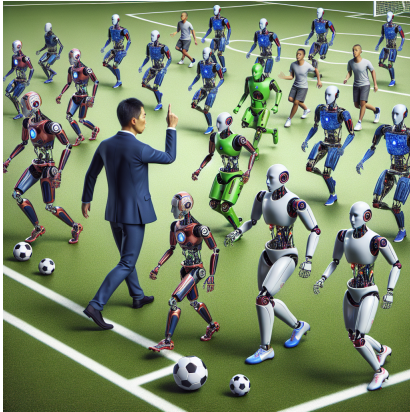


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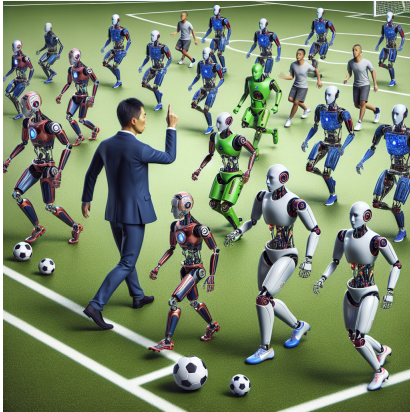


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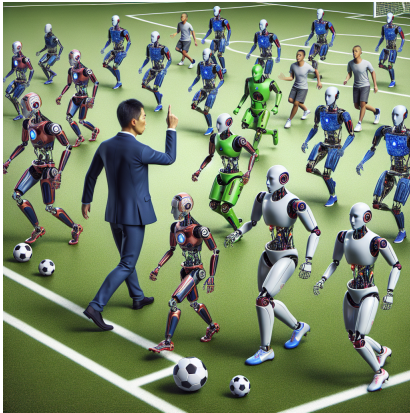


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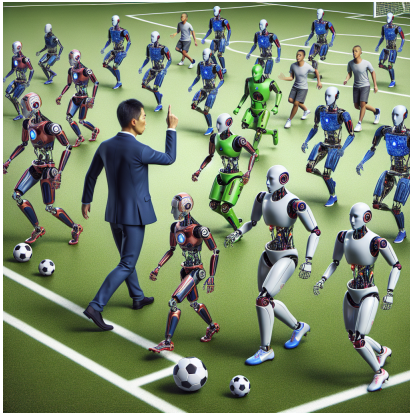


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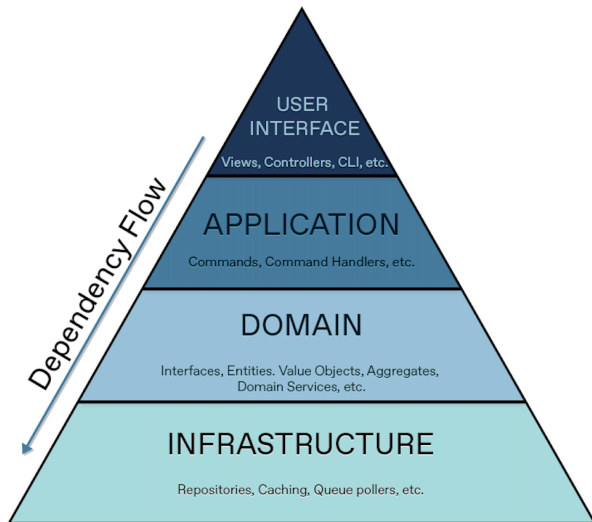


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Handwritten notes: "begin" (in red) and "services" (in green) with an arrow pointing to the Application Layer. "data repositories" (in blue) is written below the Infrastructure Layer.



Business Logic in Layers



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Software Development Life Cycle (SDLC)

- The **SDLC** is a **framework** that describes the stages involved in **developing software applications**.
 - It includes phases like **planning**, **analysis**, **design**, **implementation**, **testing**, and **maintenance**.
 - The SDLC helps ensure that software is developed in a structured and efficient manner, leading to high-quality products.
 - Software methodologies provide frameworks for planning, designing, developing, testing, and maintaining software projects.
 - They help teams manage project complexity and ensure high-quality deliverables.
- functional user need
performance business impact +

software metrics



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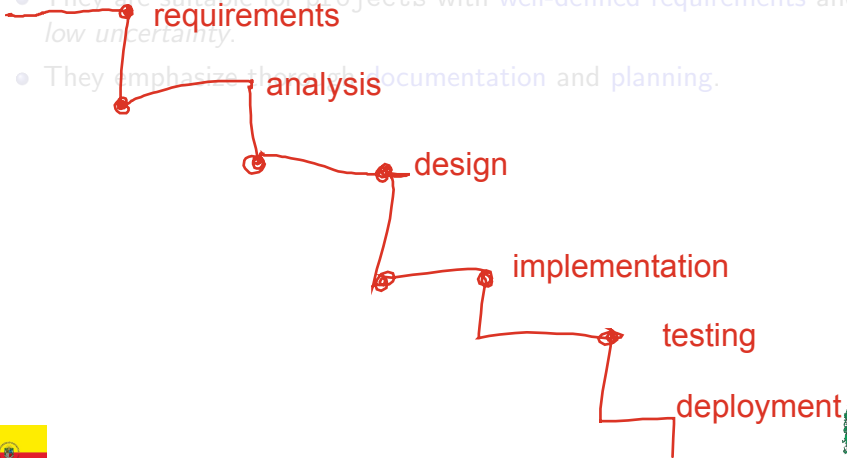


Traditional Methodologies

- **Waterfall:** A **linear approach** where each phase must be *completed before moving* to the next.

- They are suitable for projects with **well-defined requirements** and *low uncertainty*.

- They emphasize thorough **documentation** and **planning**.



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R.U.P.



Agile Methodologies

- Emphasize iterative development, customer collaboration, and flexibility.
 - They are based on the **Agile Manifesto**, which values individuals and interactions over processes and tools.
 - Examples include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
 - Agile methodologies are suitable for projects with changing requirements and high uncertainty.
 - They promote adaptive planning, evolutionary development, and early delivery of valuable software.
- Handwritten annotations:
- A red circle around "Agile Manifesto" with a red line pointing to "startups".
 - A blue circle around "Scrum" and "Kanban" with a blue line pointing to "startups".
 - A red line connecting "Money", "Quality", and "Time" to "startups".



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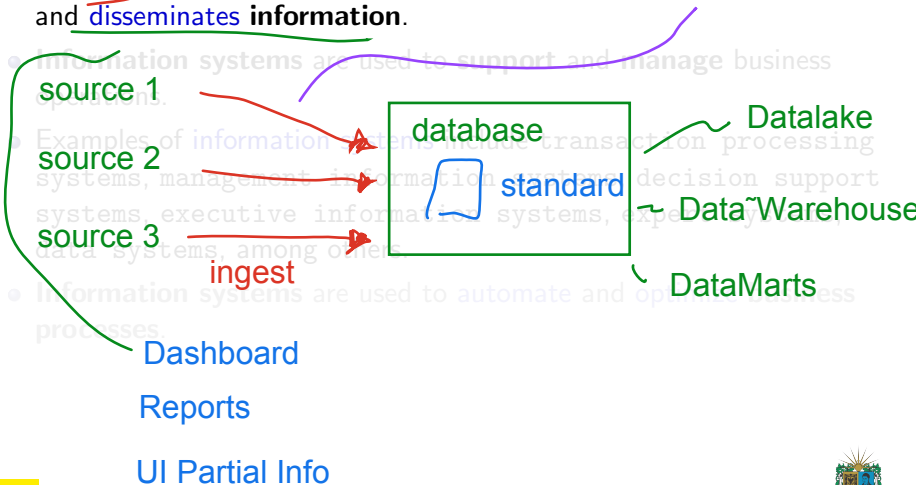
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Data Systems

- A **Data System** is a **system** that *collects, processes, stores, and retrieve data*.
- Examples of **data systems** include databases, data warehouses, data lakes, data marts, data cubes, data streams, data lakes, among others.
- **Data systems** are used to **store** and **analyze** data.



Expert Systems

- An **Expert System** is a system that uses **knowledge** and **reasoning** to solve problems.
- Examples of **expert systems** include diagnostic systems, predictive systems, prescriptive systems, decision support systems, among others.
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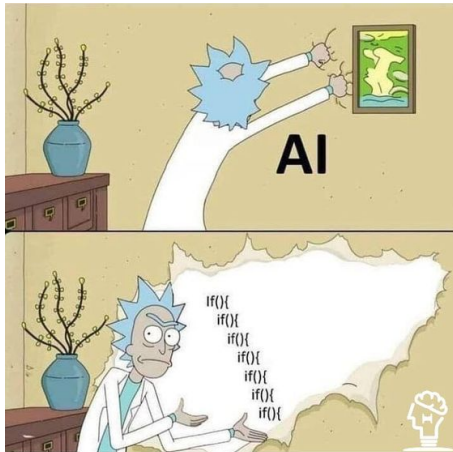
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Expert Systems as Classical Artificial Intelligence

Here there is a great example of a **diagnostic system**.



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can impact **business operations**.
- Examples of **risks and failures** include security breaches, data loss, system downtime, performance issues, compliance violations, among others.
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Questions?



www.linkedin.com/in/casierrav

